1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In bladder cancer, PROS1 expression positively correlates with MKI67 (Ki-67) expression, indicating a proliferative state. Among tumors with high TIL density, elevated PROS1 levels are associated with shorter progression-free interval (PFI) compared with tumors with low PROS1, independent of standard clinical covariates (age, sex, stage, grade). This PROS1–MKI67 axis sustains tumor proliferation despite immune cell presence, effectively decoupling immune infiltration from tumor control. Consequently, PROS1 serves as a prognostic biomarker within TIL-rich bladder tumors and represents a potential target to restore immune-mediated tumor control and prolong PFI.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PROS1, MKI67
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 is expressed in tumors and associated with the proliferation marker Ki-67, so why might tumors with apparent immune infiltration still progress rapidly? (2) I built on the single Tier 1 path linking PROS1 expression to Ki-67 (MKI67) because it directly connects a tumor-intrinsic factor to proliferative state that could negate immune control.

MKI67: MKI67) on disease progression — together these suggest PROS1-driven proliferation could uncouple immune infiltration from effective tumor control.

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis using bladder cancer samples to quantify PROS1 and MKI67 expression, estimate TIL density from histology or deconvolution, and model PFI with interaction terms to detect PROS1–MKI67 and PROS1 × TIL effects;

Prediction 2: In vitro manipulation of PROS1 in bladder cancer cell lines to assess effects on Ki-67 and proliferation;

Prediction 3: Co-culture assays to examine how PROS1 modulation influences immune cell–mediated effects;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions